

## JEREMIAH WILLIAM SEBASTIAN

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### EDUCATION

#### SINGAPORE MANAGEMENT UNIVERSITY

Jan 2026 - Apr 2027

Master of IT in Business (Data Science and Analytics Track) (CGPA:3.58/4.0)

#### Bina Nusantara UNIVERSITY – Indonesia

Aug 2021 - Aug 2025

Bachelor of Science in Computer Science (Major in Software Engineering, Minor in Mobile Development) (CGPA: 3.67/4.0)

### ACADEMIC PROJECTS

#### Customer Churn Prediction for E-Commerce

2026

- Building an end-to-end supervised ML pipeline to predict customer churn from behavioural and transactional data using Python and NumPy, with data cleaning and feature engineering completed.
- Applying predictive modelling techniques including classification algorithms to identify at-risk customer segments and surface actionable retention insights.

#### Fraud Detection Using Supervised Machine Learning Model (KNIME)

2026

- Designed custom performance indicators — Behavior Score, Risk Intensity, Average Item Value, and Velocity Ratio — to flag anomalies, mirroring real-world KPI design for monitoring business risk
- Addressed a 4.4% class imbalance in the dataset using SMOTE, ensuring reliable and consistent model outputs
- Evaluated multiple predictive models, identifying the best performer with 92.3% recall, successfully catching 12 of 13 risk cases

#### Statistical Reassessment of Value Drivers for HDB In POST-COVID Singapore (SQL, Python, 70K rows)

2026

- Conducted data analysis to quantify how an accessibility-based "Convenience Score" drove a S\$183K valuation gap, translating statistical findings into a clear business narrative for faculty review
- Applied ANOVA, Tukey HSD, correlation, and regression techniques to test and validate which value drivers most influenced market outcomes
- Presented methodology and findings to faculty, demonstrating ability to turn complex analysis into actionable, stakeholder-ready insights

#### Query Processing Optimization

2026

- Automated extraction of birth-month data from raw date fields via a STORED Generated Column strategy, removing manual calculation overhead while ensuring 100% data accuracy
- Redesigned non-SARGable queries into indexed, SARGable predicates, shifting execution from full table scans to efficient index range scans
- Achieved a ~15x reduction in logical reads and ~70% reduction in query cost (208,580 → 62,686), demonstrating scalable data-processing efficiency

### EXPERIENCE

#### Wisely.id – Indonesia (<https://www.wisely.id>)

Feb 2025 – Nov 2025

#### Founder and Full Stack Engineer

- Built a business performance dashboard that converted transactional data into trend reports, enabling owners to identify high-turnover products and inform forward-looking purchasing decisions
- Designed automated alerts to flag inventory risk ahead of time, improving timeliness of operational reporting and reducing reactive decision-making
- Maintained the underlying data structure to ensure accuracy and consistency of reporting across the business

#### Bank Central Asia – Jakarta, Indonesia

Feb 2024 - Feb 2025

#### Software Engineer Intern

- Engineered scalable backend systems using Spring Boot, designing and optimizing RESTful APIs supporting core application features
- Supported a cross-functional migration project, coordinating with senior engineers to modernize a core system while maintaining business continuity
- Contributed to process improvements that enhanced system reliability and operational efficiency

### ADDITIONAL

- **Technical Skills:** Python, SQL, PostgreSQL, Power BI, KNIME, Machine Learning, Statistical Analysis, Git, AI-Assisted Development (Claude Code), Microsoft Excel (Advanced)
- **Language Skills:** English (Professional — IELTS 7.5), Indonesian (Native), Mandarin (Elementary)
- **Work Authorization:** Singapore (Student Pass), Indonesia (Citizen)